Anjali Parashar

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Research Statement

Testing & Evaluation of autonomous systems using sampling based methods, and Bayesian Experimental Design (BED) for limited budget testing, with applications to data mixing for VLMs/VLAs and related work on optimization theory. Looking for full-time roles starting May/June 2027 in robotics and autonomous systems.

Education

- **Massachusetts Institute of Technology** — Advisor: Prof. Chuchu Fan
  PhD in Robotics, Reliable Autonomous Systems Lab (GPA 4.9/5) — *June 2023 - On-going*
- **Massachusetts Institute of Technology** — Advisor: Dr. Anuradha Annaswamy
  M.S. in Mechancial Engineering, Active Adaptive Control Lab (GPA 4.9/5) — *Sept 2021 - May 2023*
- **Indian Institute of Technology, Indore** — Advisor: Prof. Santhakumar Mohan
  Bachelor of Technology in Mechanical Engineering (GPA 9.3/10) — *August 2016 - June 2020*

Work Experience

- **NVIDIA Autonomous Systems and Physical AI Research (ASPIRE) Group** — Santa Clara, USA
  Research Intern — *June 2026-Ongoing*
  - **Data-mixing for VLMs**: Worked in **Safety Evaluation and Assurance** team on developing algorithm for data mixing for fine-tuning VLMs and VLAs for Physical AI applications using Bayesian modeling and bilevel optimization.
- **John Deere Pvt. Ltd. (Technology Center India)** — Remote, India
  Graduate Engineering Trainee — *August 2020 - February 2021*
  - **Mobile robot development**: Worked on the design and development of mobile robots, using ROS for mathematical modelling, mapping, localization & navigation, hardware interface, and remote control of AMR integrated with ROS

Research Experience

- **PhD Thesis (June 2023-ongoing, MIT)**: Developing benchmarking and evaluation tools for cost-aware evaluation and failure discovery of autonomous systems using Bayesian Experimental Design (BED), and accelerated MCMC sampling, with qualitative investigation of failures using foundation models. Work presented at ICRA 2025 [Website], NeurIPS Bayesian Decision Making and Uncertainty (BDU) Workshop 2024, Allerton Conference 2025, [Website], CoRL 2025 [Website], ICLR 2026 [Website], NeuS 2026, CoRL 2026 [arXiv] (accepted, academic collaboration with NVIDIA) and NeurIPS 2026 (accepted).
- **M.S. Thesis (Sept 2021-May 2023, MIT)**: Stability analysis and implementation of accelerated algorithms for solving constrained convex optimization problems with an application in optimal power flow (OPF) problems. Work presented at CDC 2022, ACC 2023 [Thesis PDF]. Related work published in TMLR 2025, and ICLR 2026 [PDF] (J2C certification).
- **B.Tech Thesis (August 2018-December 2019, IIT Indore)**: Fault tolerance control for underactuated Autonomous Underwater Vehicles (AUVs), verified robustness of controller to oceanic currents, validated performance using MATLAB simulations [PDF].

Awards & Features

- Work on SEED-SET, Experimental design for autonomous system testing featured in MIT News [link]
- Recognized as Top Reviewer by NeurIPS 2026, awarded to top 8% of highest-scoring reviewers
- Recipient of IEEE Student Travel Support Grant for CDC 2022, CoRL 2025 Travel Grant, CoRL 2026 Travel Grant
- Recipient of Siemens Fellowship for Master's Thesis (2021-22), NTSE Scholarship, India (2014-2020, top 0.1%)

Selected Publications

- **NeurIPS (2026), Accepted**: A. Parashar, C. Sobolewski, Y. Li, F. Chen, C. Fan, "Adaptive Test Case Discovery for LLM-Assisted Decision Making in High-Stakes Domains"
- **CoRL (2026), Accepted**: A. Parashar, R. Luo, A. Sharma, S. Veer, E. Schmerling, C. Sobolewski, M. Yu, C. Fan, M. Pavone, "Coverage Aware Active Evaluation for Failure Discovery with Paired Systems"[arXiv pre-print]
- **ICLR (2026)**: A. Parashar, Y. Li, E. Y. Yu, F. Chen, J. Neidhoefer, D. Upadhyay, C. Fan, "SEED-SET: Scalable Evolving Experimental Design for System-level Ethical Testing" [PDF]
- **CoRL (2025)**: A. Parashar, J. Zhang, Y. Li, C. Fan,"Cost-aware Discovery of Contextual Failures using Bayesian Active Learning" [PDF]
- **NeurIPS BDU Workshop (2024), Allerton Conference (2025)**: A. Parashar, J. Zhang, K. Garg, C. Fan, "Failure Prediction from Limited Hardware Demonstrations" [Website]
- **TRO (2025)**: C. Dawson, A. Parashar, C. Fan, "RADIUM: Predicting and Repairing End-to-End Robot Failures using Gradient-Accelerated Sampling" [PDF]
- **RA-L (2024), ICRA (2025)**: A. Parashar, J. Yin, C. Dawson, P. Tsiotras, C. Fan, "Learning-based Bayesian Inference for Testing of Autonomous Systems" Website

Skills Summary

- **Courses**: Bayesian inference and modelling, Computational Cognitive Science, Machine Learning, Real Analysis, Stochastic systems, Nonlinear Control, Identification, Estimation & Learning.
- **Software & Hardware**: Python, MATLAB, ROS, CARLA, Webots, F1-Tenth, UR3E Collaborative Robot Arm, Turtlebot